48V Electrical Architecture

Document role

- This file owns the final `48V` electrical architecture for the active build baseline.
- Use it for house-bank structure, alternator-path direction, shutdown logic, and manual control-path intent.
- Keep full conductor schedules, fuse matrices, and layout-level implementation detail in `docs/implementation/`.
- Keep unresolved vendor gates, risk state, and follow-up closure items in `docs/core/TRACKING.md`.
- Keep broad project sequencing and day-to-day execution framing in `docs/core/PROJECT.md` or the active plan docs.

As-of date: `2026-08-12`

Purpose: hold the finalized, concise `48V` house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- Electrical topology diagram
- Electrical fuse schedule
- Mechman / WS500 / APM-48 install guide
- Systems
- Tracking
- Estimated BOM

Final direction

- House architecture stays `48V` core with `3x 51.2V 100Ah` batteries in parallel.
- Alternator charging is the dedicated `48V` secondary alternator path; obsolete pre-Mechman charger hardware is removed from active planning.
- Protection baseline is `Mechman + WS500 + Balmar APM-48`.
- Manual alternator-charge enable/disable is through Ford `Upfitter Switch #3`.
- `Upfitter #3` is a low-current control signal only. It does not carry alternator output current.
- `WS500` white `Feature-In` is reserved for future fault-interlock work, not required in Phase 1.

Current commissioning state (`2026-09-02`)

- `48V` bus has been live-tested: owner measured `55.5V` throughout the system, including at the MultiPlus.
- MultiPlus-II DC/inverter mode has been switched on with inverter light illuminated, slight normal hum, and no reported error lights.
- SmartShunt and Orion-Tr Smart are visible in VictronConnect.
- Cerbo GX access point/remote-console workflow is active; Cerbo power is a small inline fused feed from the `48V` system side and MultiPlus communication is via `VE.Bus` RJ45.
- Shore charging has been tested through the MultiPlus at household-outlet current limits. The first short test proved basic AC-in/charger function; later owner verification redid the settings and confirmed first-battery behavior entered bulk, then quickly transitioned to absorption at/near `100%` as planned.
- MultiPlus LiFePO4 charge-profile programming/verification is treated as closed for the current shore-charger setup. Current target is `56.8V` absorption/charge, `54.0V` float, short absorption dwell, equalization off, conservative LiFePO4 storage behavior, and source-current limit matched to the actual shore circuit. The Dumfume manual's `58.4V +/-0.2V` value is documented, but because the same manual also lists `58.4V` as charge-limit/over-charge protection voltage, it is not the active routine charger target. `DVCC` remains disabled unless a documented BMS/GX control path is added.
- The electrical module is hard-mounted and tied into the hard-mounted Bench/Galley structure. Owner reports the batteries and ICECO are positively restrained; the not-yet-installed air fryer, bench lid, monitor/laptop package, and other storage remain separate retention work.
- All three batteries' `2/0 AWG` branch leads are complete, and the `3x 48V` bank has been permanently paralleled and in regular use for an extended period. The earlier Battery 2/3 isolated-charge staging language is historical and no longer an active prerequisite.
- Owner reports regular use of the `12V`, `48V`, and `120V` systems. All four duplex outlet boxes and the other `120V` enclosures are complete. Remaining AC safety verification is explicit GFCI `LINE/LOAD`/downstream trip behavior, polarity, PE continuity, neutral-isolation behavior, and the required MultiPlus-to-chassis plus shell bonds—not another enclosure build.
- Solar pass-through, disconnect, fixed wiring, and MPPT landing are complete; no panels are connected. The secondary alternator remains disconnected, with the protected PH-VAN red lead and Wakespeed read/save/programming preceding controlled first charge.

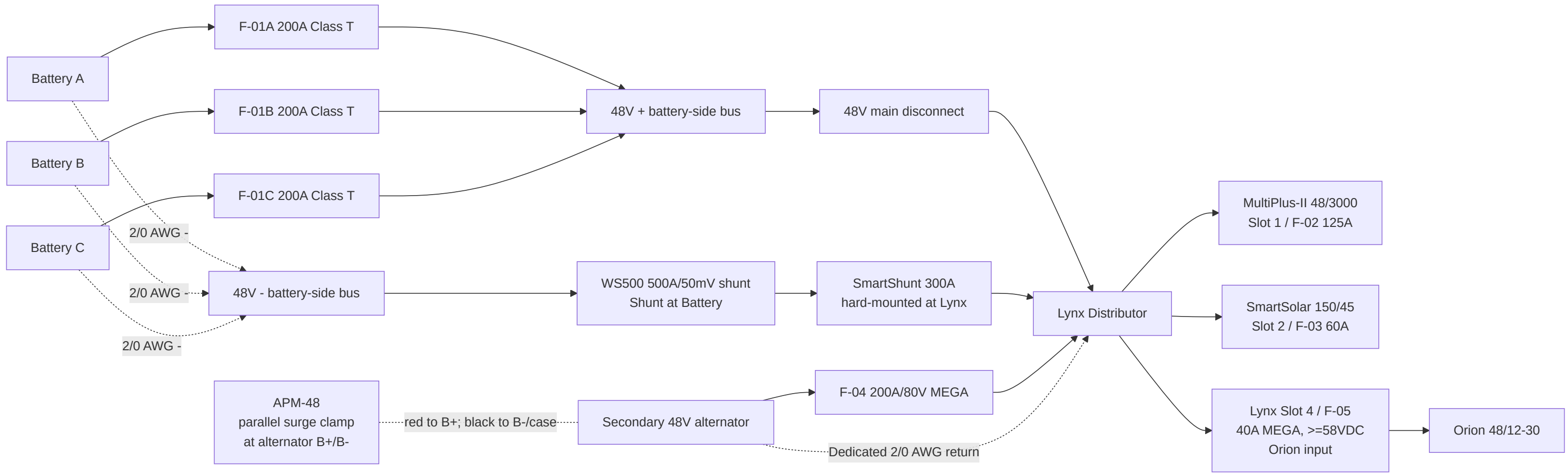
Locked component set

Function	Locked baseline	BOM row(s)
House batteries	3x Dumfume 51.2V 100Ah LiFePO4	3
Main house disconnect	Victron 275A battery switch	5
Main fused distribution	Victron Lynx Distributor M10	6
Battery branch protection	F-01A/B/C 200A Class T; manual-backed battery limit is 200A max continuous discharge per battery, with terminal/manufacturer fuse guidance still not separately specified; owner confirmed 3 holders and 4 slow-blow Class T fuses total	7
Inverter/charger	MultiPlus-II 48/3000/35-50	12
48V to 12V charger	Orion-Tr Smart 48/12-30; fed directly from Lynx Slot 4 through one verified 40A MEGA (58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) into the existing 6 AWG input pair; no separate inline/DIN input fuse; keep 6 AWG and F-07 60A/80V on 12V output	20, 10, 11
Monitoring	Cerbo GX + SmartShunt 300A	22, 23
Solar controller	Purchased Victron SmartSolar MPPT 150/45; retain it for the final array	25
Solar array	Purchased 4x Renogy 175W flexible monocrystalline = 700W; current candidate is one 4S string at 78.0V Vmp, 95.6V Voc, 8.98A Imp, and 9.50A Isc. Published -0.31%/C Voc coefficient gives 114.86V nominal calculated Voc at -40C. Exact received labels and hot-start/tracking behavior remain acceptance gates.	24
Alternator kit	Mechman 48V secondary alternator kit with WS500	168
Load-dump clamp	Balmar APM-48	169
Alternator branch fuse	F-04 200A/80V MEGA in Lynx Slot 3	170
WS500 PH-VAN and shunt sensing	One Eaton/Bussmann HEB-AA in-line holder + Littelfuse KLKD015.T (15A, 600VAC/DC) midget fuse on the short red combined regulator-power/positive-sense lead; separate 500A/50mV Wakespeed shunt on the battery side of the hard-mounted Victron SmartShunt in the common battery-negative path	171
WS500 Upfitter #3 control kit	F-15 + control wire + holder/terminals	176

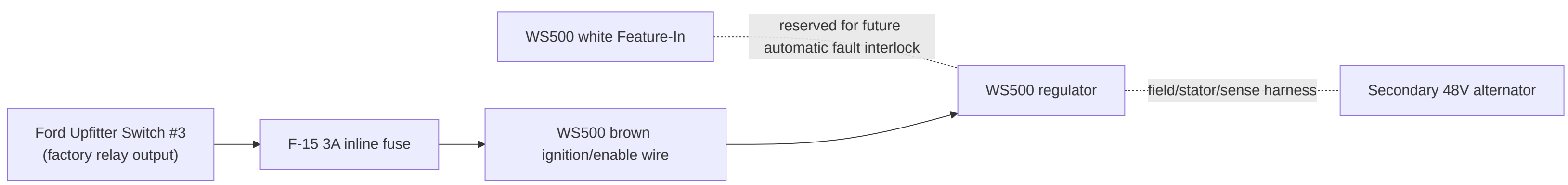
Battery manual limits and charger-programming baseline

- Battery model basis: Dumfume `51.2V 100Ah` LiFePO4, `3x` in parallel (`1S3P`; manual allows up to `1S4P`).
- Per-battery charge voltage: `58.4V +/-0.2V`; charge-limit/over-charge protection voltage also listed as `58.4V`.
- Per-battery current references: recommended charge `20A`; max continuous charge `100A`; recommended discharge `50A`; max continuous discharge `200A`.
- Bank-level reference for current `3x` setup: recommended charge `60A`; max continuous charge `300A`; recommended discharge `150A`; max continuous discharge `600A`.
- Protection thresholds: over-discharge protection `36.8V`, recovery `43.2V`; discharge overcurrent protection `600A`; short-circuit protection `1800A`.
- Temperature thresholds: low-temp charge protection approx `41F-50F` with recovery at `50F`; high-temp protection listed as `157F` / `140F`.
- MultiPlus-II charger limit is `35A`, so full-output charging is below the bank's `60A` recommended-charge reference (`~11.7A` per battery in `3P`). For first household-outlet tests, keep AC input current at `10A`, then `12A` max on a normal `15A` source. Current charger-programming target is `56.8V` absorption/charge, `54.0V` float, `52.8V` storage if shown, `0.5h` max absorption, equalization off, and temperature compensation off.

48V power path



Alternator control path



Control logic

- `Upfitter #3 ON` : `WS500` brown wire is energized, regulator is allowed to run, alternator charging can occur.
- `Upfitter #3 OFF` : `WS500` is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use `Upfitter #3` as the manual alternator-charge shutdown control.
- Do not use the main `48V` disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why `Upfitter #3`

- It is already a factory-switched, relay-driven `12V` control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
`F-01A/B/C`	Battery branch positive protection	`200A` Class T provisional	`2/0 AWG`
`F-04`	Alternator branch into Lynx Slot 3	`200A/80V` MEGA	`2/0 AWG`
`F-05`	Lynx Slot 4 -> Orion `48V` input	MEGA `40A`, body-marked `>=58VDC`; Victron `CIP138040020 40A/80V` replacement fallback	Existing `6 AWG` direct to Orion; single input fuse
`F-06`	Retired standalone Orion input fuse position	Not installed	MIDI/FKS/DIN concepts superseded; do not stack with `F-05`
`F-12/F-13-PHVAN`	WS500 `PH-VAN` combined regulator-power / positive-voltage-sense red lead	Littelfuse `KLKD015.T`, `15A` fast-acting midget fuse, `600VAC/DC`, in Eaton/Bussmann `HEB-AA 600V` in-line holder	In-line beside `F-04` load/alternator-side stud; short `14 AWG` pigtails; sealed step-down splice to `16 AWG` harness; no panel, DIN rail, or enclosure

`CERBO-PWR`	Cerbo GX low-current power feed	`1A-3A` inline fuse/holder rated for the `48V` bank maximum; system/load side of main disconnect preferred for bench shutdown	`18 AWG` red/black duplex acceptable
WS500 current-sense pair	Purple/grey to Wakespeed shunt in common battery-negative path	No fuse; purple/high to battery-negative side, grey/low to SmartShunt/Lynx system side; twist pair and route away from noise; configure `Shunt at Battery`	harness lead
`F-15`	Upfitter #3 to WS500 brown ignition wire	`3A` inline ATO/ATC on 12V control circuit	`16 AWG` TXL/GXL control wire

Major conductors

- Battery branch and main `48V` trunk wiring: `2/0 AWG`.
- Parallel battery-current sharing target is similar **total loop resistance** per battery path, not equal positive-only length. The current bench layout may use short/medium/long positive leads balanced by long/medium/short negative leads; do not add unnecessary cable coils solely to make positive leads identical.
- Secondary alternator positive path (`ALT B+ -> F-04 -> Lynx`): `2/0 AWG`, `~20 ft` one-way planning basis. The `APM-48` is wired in parallel across alternator `B+` and `B-`/case; it is not in series with this conductor.
- Secondary alternator dedicated negative return (`ALT B-/case -> long 2/0 -> Lynx -`): `2/0 AWG`, `~20 ft` one-way planning basis. Do not use sheet metal/chassis as the normal return.
- Common battery-negative measurement path: `battery negative combine -> Wakespeed 500A/50mV shunt -> Victron SmartShunt -> Lynx/system negative`. The SmartShunt remains hard-attached to the Lynx. Configure `Shunt at Battery`; purple/high faces battery negative and grey/low faces the SmartShunt/Lynx system side.
- Orion `48V` feeder: existing `6 AWG` remains as the no-rework path from Lynx Slot 4 and is protected by `F-05 40A` MEGA (`>=58VDC`); it is electrically overkill, and flexible fine-strand `10 AWG` would be adequate if ever replaced for unrelated reasons. MPPT battery leads and Orion `12V` output remain `6 AWG`.
- Upfitter #3 control lead to WS500 brown wire: `16 AWG` TXL/GXL planning basis, `~6 ft` one-way assumed until measured.

Grounding and bonding separation

- **AC protective earth:** the MultiPlus external `M6 PE` lug is mandatory in this mobile installation. Bond it with at least the Victron-manual `4 mm²` conductor; current build default is `10 AWG` green stranded copper to a verified truck-chassis bond point. Do not use the aluminum camper shell or 80/20 as the only protective-earth conductor.
- **Exposed aluminum shell:** add a separate corrosion-compatible bonding jumper from the Hiatus shell to the chassis/equipment-ground network and verify low-resistance continuity. Mechanical shell/bed mounting contact is not accepted as proof of a durable bond.
- **AC neutral:** do not add a fixed neutral-ground bond downstream. The MultiPlus internal ground relay remains enabled: AC-out neutral bonds to chassis in inverter mode and that bond opens when external AC is accepted.
- **DC negatives:** do not jumper MultiPlus PE/case to Lynx negative or the `12V` negative bus. Keep normal `48V`/`12V` current on dedicated returns. The Mechman branch uses a dedicated `2/0` negative directly to Lynx/system negative. Both monitoring shunts remain in series in the one common battery-negative path, so all battery current must pass through `battery -> Wakespeed shunt -> SmartShunt -> Lynx/system` regardless of alternator case/chassis behavior. The SmartShunt remains directly attached to the Lynx; prove no return is landed on the battery side of either shunt before charging.

APM-48 wiring intent

- Mount the `APM-48` at the alternator end of the `48V` branch, as close to the alternator output/ground points as practical.
- Treat the APM-48 as a parallel surge/load-dump clamp across the alternator, not a series device in the alternator charge-current path.
- Connect APM red to alternator `B+`.
- Connect APM black to alternator `B-`, or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do **not** place either APM connector under the main battery cable lugs; Balmar's quick-start sheet explicitly prohibits placing APM connectors under the battery cable lugs.

Normal operating sequence

1. Main `48V` disconnect closed.
2. Engine running.
3. `Upfitter #3` switched `ON`.
4. `WS500` enabled and regulating.
5. Alternator charges the house bank through `APM-48` and `F-04`.

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch `Upfitter #3` `OFF` to disable the `WS500`.
2. Wait for alternator charge current to collapse.
3. Open the main `48V` disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The `APM-48` is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the `48V` bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.
- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- `F-04` `200A/80V` MEGA stock and the Eaton/Bussmann `HEB-AA` + Littelfuse `KLKD015.T` in-line `PH-VAN` branch package were ordered `2026-08-09`; receipt, body-marking, physical fit, and small termination-stock checks remain open. Procurement/physical inventory confirmation for the separate `F-15` `3A` Upfitter-control hardware also remains open.
- Physical fit, stud-size, and jumper-length confirmation for the Wakespeed shunt inserted on the battery side of the SmartShunt in the common battery-negative path; the SmartShunt-to-Lynx connection does not move.
- Exact Mechman field-voltage/WS500 derate setting and alternator negative/case-isolation behavior in the installed kit.
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the `WS500`.
- Whether future automatic fault-interlock logic should be added on the WS500 `Feature-In` or through an external relay.
- Solar purchase release: final MaxxAir/track survey, `1:1` panel/Starlink templates, two-panel cassette design, preliminary itemized `<=75 lb` budget, and any required insurer/Hiatus acceptance. Product/topology are locked; purchase is not yet released.
- Solar installation/travel release: received labels/polarity, built-cassette load/uplift proof, actual complete `<=75 lb` weigh-in, branch/disconnect/cable/umbilical/entry hardware, bond-coupon/removal proof, estimated hot-string voltage minus measured route drop, and stationary hot-roof commissioning.

Rule for future edits

- Update this file first for any `48V` architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.